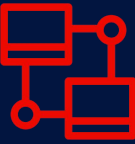


Strategy for Implementing Generative AI in Business

CSIT205 Generative AI – 2025



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Wollongong
Dr. Rory Sie
(rorys@uow.edu.au)

*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*

Dr Andrew Yoong, SIM



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, with the 'U' at the top, 'O' in the middle, and 'W' at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
community.

Find out more, seek support or join
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Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

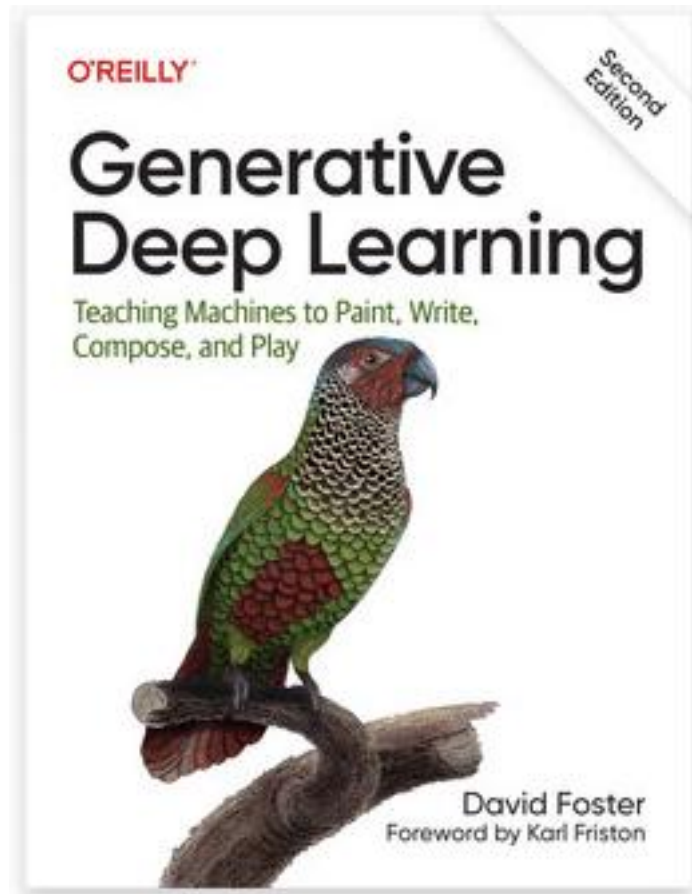
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

What did we discuss last week?

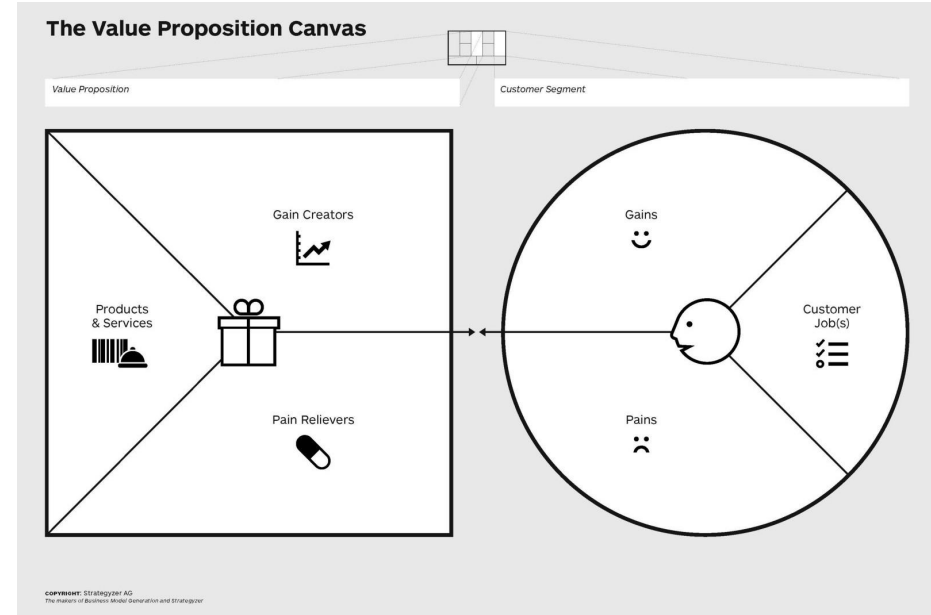


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Value Proposition Canvas

Shows how your tool can add value

- Client profile:
 - Jobs: tasks, needs
 - Pains: risks, costs, challenges, etc.
 - Gains: wishes, expectations
- Value proposition:
 - Products & Services: thing that does the job
 - Gain creators: e.g. increased productivity
 - Pain relievers: e.g. cost / time savings

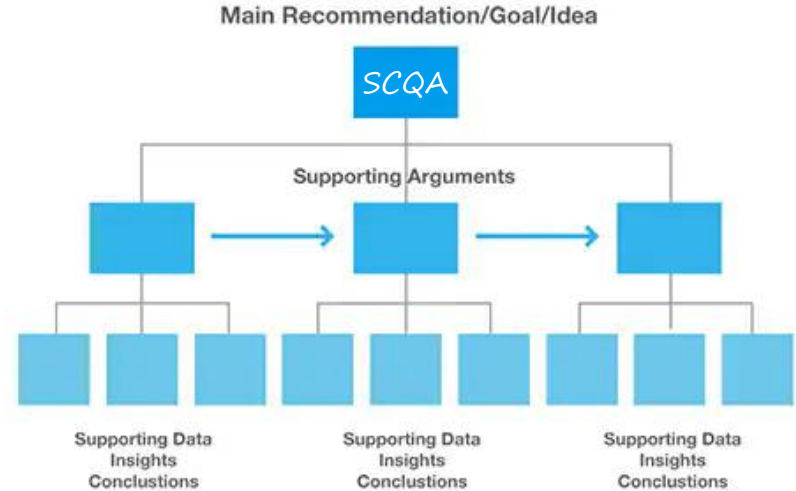


Strategyzer AG

The SCQA method

Situation, Complication, Question, Answer

- Created by Barbara Minto (McKinsey)
- Part of the Pyramid Principle
- Excellent way to structure your argument
- Situation: What is the status quo, the current state?
- Complication: What challenges do you face?
- Question: What questions arise to address the complication?
- Answer: How you address the complication



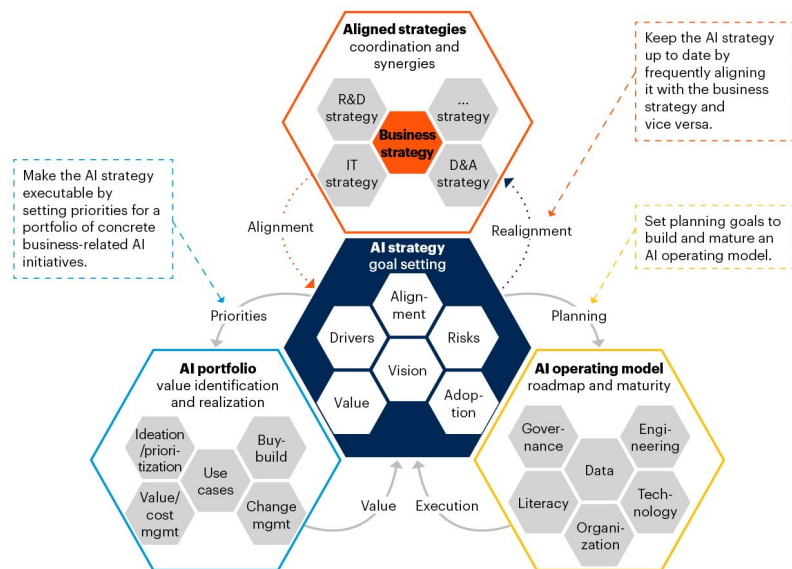
<https://www.mckinsey.com/alumni/news-and-events/global-news/alumni-news/barbara-minto-mece-i-invented-it-so-i-get-to-say-how-to-improve-it>

The Vision in Context

It is pivotal to your Strategy and is closely related to:

- *Value: What are the company's priorities and what use cases do we have?*
- *Drivers: what is driving the need / urgency for Generative AI?*
- *Alignment: a GenAI vision should follow the business vision.*
- *Risks: Can we use GenAI in an ethical, responsible manner? How do we mitigate risks?*
- *Adoption: Which capabilities do we need to ensure GenAI is adopted?*

The AI Strategy in Context



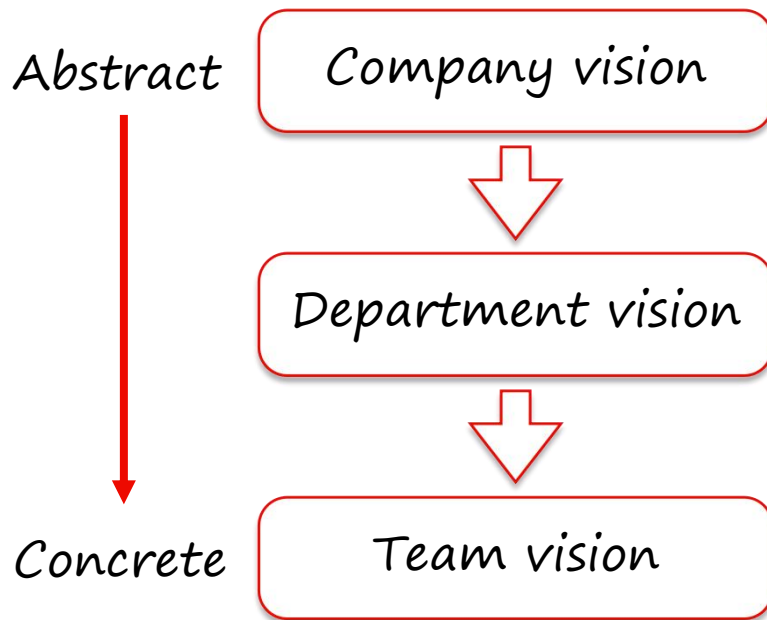
Source: Gartner
805859_C

Mission Statement

- Why are we here? What is our purpose? What are our values?
- What do we currently do as a business? What is our product or service offering?
- Who are you doing this for?
- Examples:
 - "We're on a mission to..."
 - "Our purpose is to..."

Vision Statement

- Focus on the Future: Where do we want to be in 1, 2, 5, or 10 years from now?
- You can have multiple vision statements, depending on your context
 - Team-level: 1-2 years
 - Department: 2-5 years
 - Company: 5-10 years
- A vision trickles down from the top and gradually gives more meaning to a potentially abstract vision.
- Often includes 2-4 strategic pillars of focus.
- A vision ~~can~~ will change along the way!



Assessment 2: let's go!

~~1. Form a group of 5-6 within your Lab C group. Note: max. 5 groups per lab.~~

~~2. Schedule a 'Chat with the CTO' in week 10.~~

1. Given what you have learnt last week, try to collect information from the CTO about RetailTech Innovators' vision and strategy.

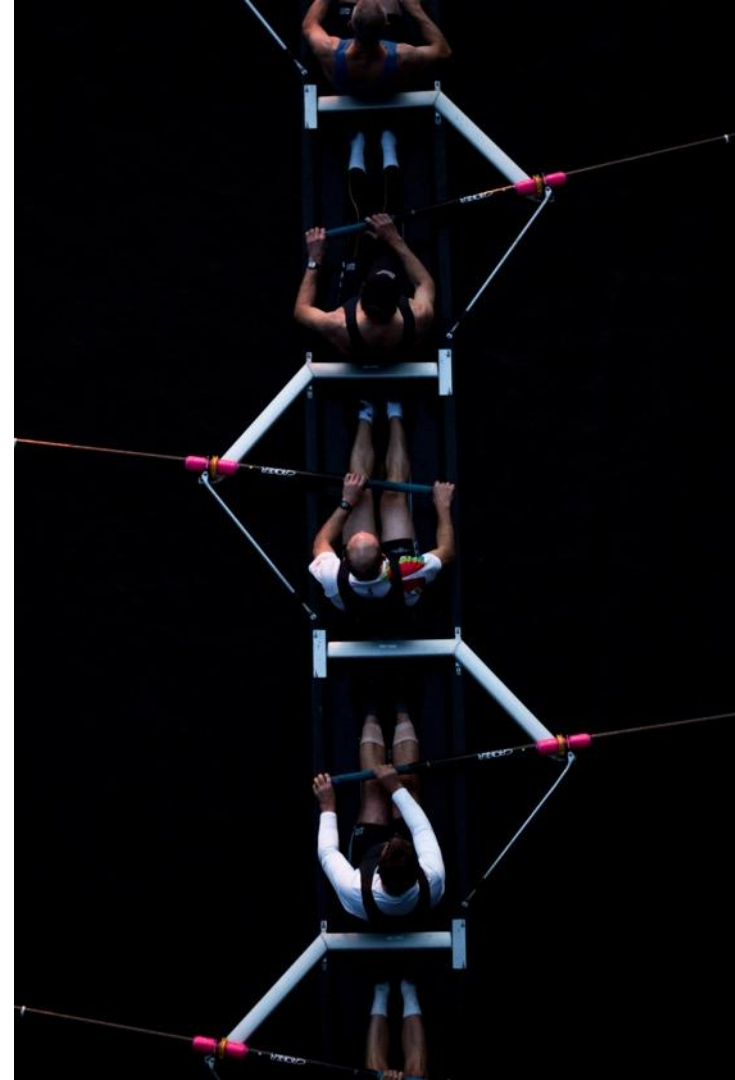
2. This will help you create your own vision and strategy for GenAI at RetailTech Innovators.

Why are we here?

CSIT205 Generative AI - Module 3

Strategy for Implementing Generative AI in Business

- AI Strategy
- AI Portfolio
- AI Operating Model (Part 1)



AI Strategy



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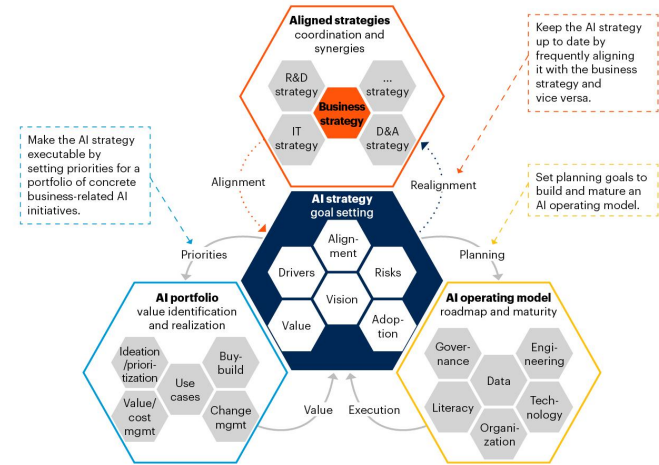
The AI Strategy

Owned by Head of AI / Data depending on company size and responsibilities

Interfaces with:

- *Aligned Strategies: continuous re-alignment*
- *AI Portfolio: generates value and receives priorities*
- *Operating Model: receives planning and executes AI Strategy*

The AI Strategy in Context



Source: Gartner
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Gartner

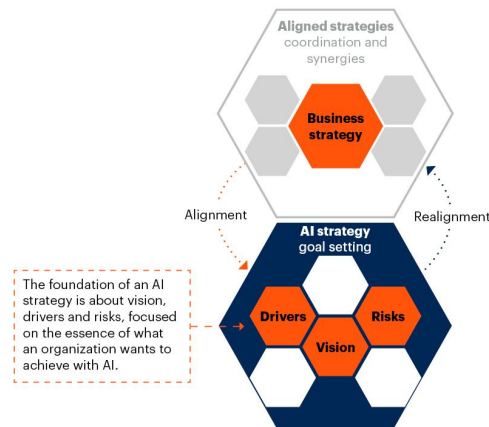
Den Hamer & Bhati, 2024

Alignment with the Business Strategy

To ensure you are working towards the same goals, with the same context

- *Vision: the AI Vision for your company (the 'how') should be aligned with the business vision, which could mention the role of AI. Owned by the*
- *Drivers: what is driving the need / urgency for Generative AI? What are the business and technology trends driving this vision (e.g. Customer Service efficiency and Agentic AI, resp.)?*
- *Risks: Can we use GenAI in an ethical, responsible manner? How do we stay in control? How do we mitigate risks? Who is mandated to make decisions?*

Core Elements of an AI Strategy



Source: Gartner
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Gartner

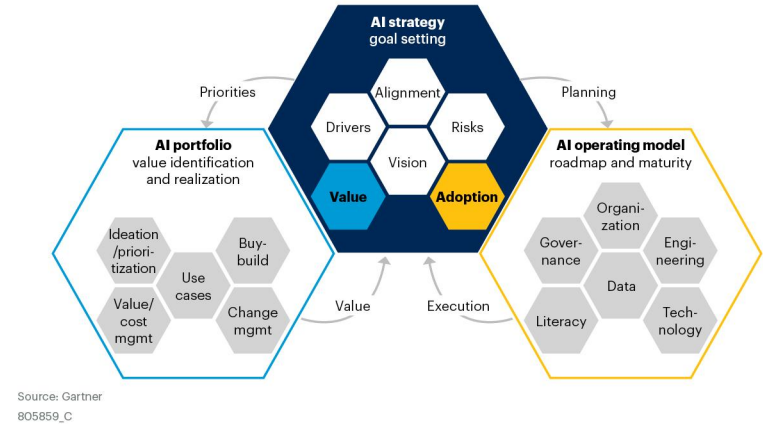
Den Hamer & Bhati, 2024

"From Strategy to Execution"

Interfacing with AI Portfolio and AI Operating Model

- **Value:** The AI Strategy communicates value priorities that inform use cases. For instance, will you start with operational efficiencies (e.g. AI assisted coding), or customer-facing solutions (e.g. personalised recommendations)?
 - How will use cases affect Key Performance Indicators (KPIs) for the business? How to measure them (i.e. metrics)?
- **Adoption:** Your adoption planning goals inform the capabilities in the AI Operating Model. They work towards KPIs to measure adoption progress.

Relationships Among an AI Strategy, an AI Portfolio and an AI Operating Model



Gartner

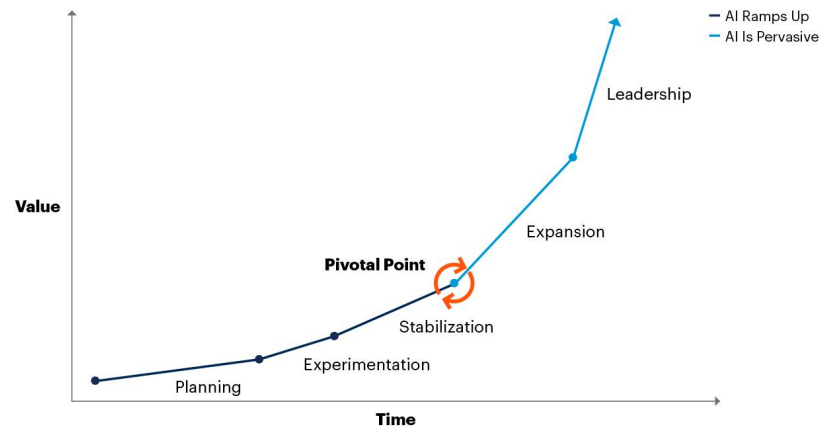
Den Hamer & Bhati, 2024

Adoption

5 Phases

- **Planning:** Mostly research into AI algorithms, potential applications, available data, etc.
 - Build a 'Guiding Coalition' (Kotter, 1996): a group of influential champions (e.g. volunteers) that can achieve short-term wins and remove barriers.
- **Experimentation:** First proof-of-concepts (PoCs) with AI. Try to show value rather than technology.
- **Stabilisation:** First AI project in production, with executive sponsor and budget for AI. Focus on sharing best practices, knowledge and skills.
- **Expansion:** Existing teams scale up and AI scales out to the rest of the company. "New roles, such as MLOps engineer, AI architect, AI validator, UX designer and AI developer, are the new normal" (Sicular et al., 2023).
- **Leadership:** All new initiatives consider AI, because the org is AI literate. Your AI experts are driving change, not just internally, but also as part of the wider community.

AI Value Curve



Source: Gartner
802232_C

Sicular et al., 2023

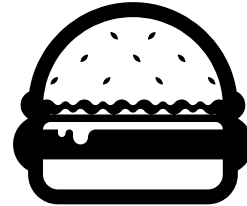
Gartner

10-minute Break!

This will help you learn



10 minutes



What are the key AI Strategy dimensions that need to align with business strategy? Select all that apply.



- ✓ 1. Value
- ✓ 2. Vision
- ✓ 3. Drivers
- ✓ 4. Risks
- 5. Adoption

The key AI Strategy dimension that interfaces with the AI Portfolio is:



A. Vision

B. Value

C. Adoption

D. Drivers

The key AI Strategy dimension that interfaces with the AI Operating Model is:



A. Vision

B. Value

C. Adoption

D. Drivers

AI Portfolio

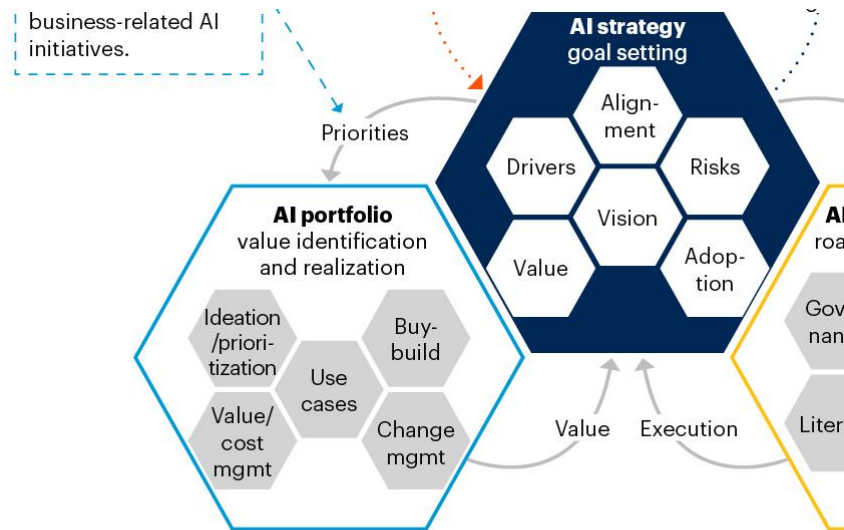


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The AI Portfolio in Context

It determines what you do, and in which order

- *Use Cases*
- *Ideation / prioritisation*
- *Change management*
- *Buy vs Build*
- *Value / cost management*



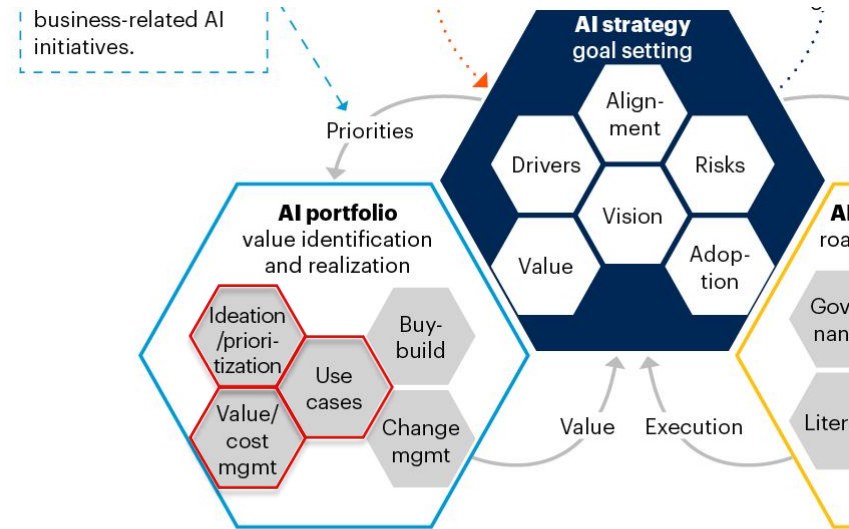
Source: Gartner
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Den Hamer & Bhati, 2022

Use Cases

Central to the AI Portfolio

- Usually managed by product / portfolio owners
- Ideation / prioritisation: What are the projects that can contribute to the business strategy and associated strategies like the Data & Analytics strategy? And what comes first?
- Value / cost management: What value will this bring us? And what will it cost us (e.g. resources, talent, tech, data acquisition)?



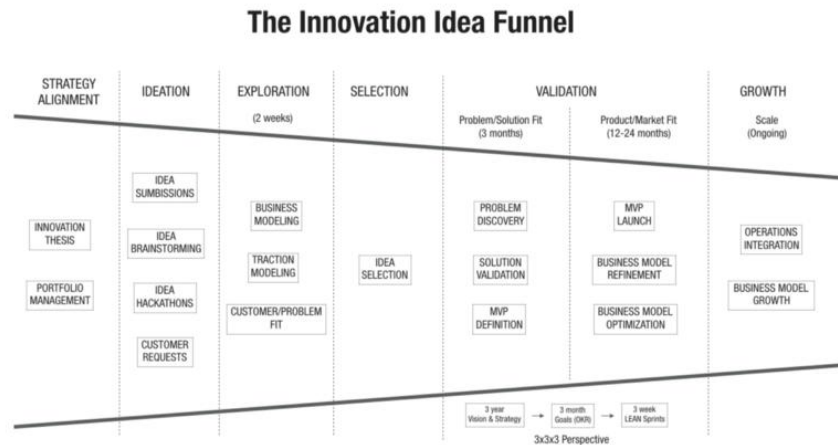
Source: Gartner
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Den Hamer & Bhati, 202

Ideation / Prioritisation

Strategies to determine priorities

- Should be aligned with strategy / vision.
- Ways to experiment:
 - Proof-of-Concepts (PoCs) (weeks)
 - Spikes (2-5 days)
 - Hackathons (1-2 days)
- An 'Idea funnel' can help select suitable projects.



Weighted Shortest Job First (WSJF)

Agile Strategy to determine priorities

- Part of the Scaled Agile Framework (SAFe).
- $WSJF = \text{Cost of Delay} / \text{Job Size}$
- $\text{Cost of Delay} = \text{Value} + \text{Time Criticality} + \text{Risk Reduction} / \text{Opportunity Enablement}$
- Tips:
 - Fibonacci numbers (i.e. 1, 2, 3, 5, 8, 13, 21) work well (try to calibrate across projects).
 - 'Person months' (PM) work well for Job Size
 - Any job of 8+ PMs should be split up.

Project / task	Value	Time Criticality	Risk Red. / Opp. Enablement	Job Size (PM)	Total
AI Assisted coding	5	3	8	2	8
Personalised recommendations	21	8	5	5	6.8
Customer service chatbot	8	8	13	5	5.8

Rationale:

- AI assisted coding is easy to implement and can save time / money (i.e. quick win)
- We already have a form of Personalised recommendations, but we can potentially earn a lot more.
- A Customer service chatbot is new will save time / money structurally, but limited.

MoSCoW method

Used within a project to determine priorities

Just because you can, does not mean you should build it.

MoSCoW:

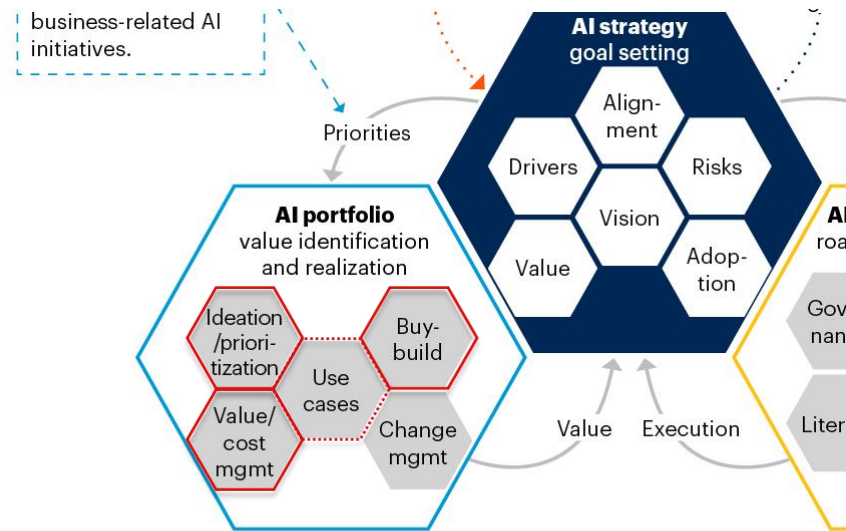
- *Must have*
- *Should have*
- *Could have*
- *Won't have*

AI coding functionality	Priority
Bootstrapping code	M
Integration with IDE	M
Automated code reviews	S
Automated deployments	C

Value / Cost management

informs your priorities, and can help with build vs. buy decisions

- Value: What are the company's priorities and what use cases do we have?
- Cost management: How much will this cost us
 - now (capital expenses; CAPEX)?
 - structurally (operational expenses; OPEX)?
- For bigger projects, a Total-Cost-of-Ownership (TCO) analysis may help determine costs.



Source: Gartner
805859 C

Den Hamer & Bhati, 2022

Build vs Buy

Cost is pivotal, but there are other considerations as well.

- When to Build:

- Custom requirements: some projects have special requirements.
- New, immature tech space: for instance if you are an 'early adopter' of tech.
- Security concerns: keep control of security yourself.
- Privacy concerns: keep control of data privacy yourself.

- When to Buy:

- Limited human resources: if you lack the people.
- Security concerns: software companies may want to prevent liability and spend more money on security.
- Privacy concerns: software companies may be more specialised in handling particular (domain) data or being compliant (e.g. HIPAA).
- Immature AI organisation: If you lack the core competencies to build it or maintain/control it, buy it.

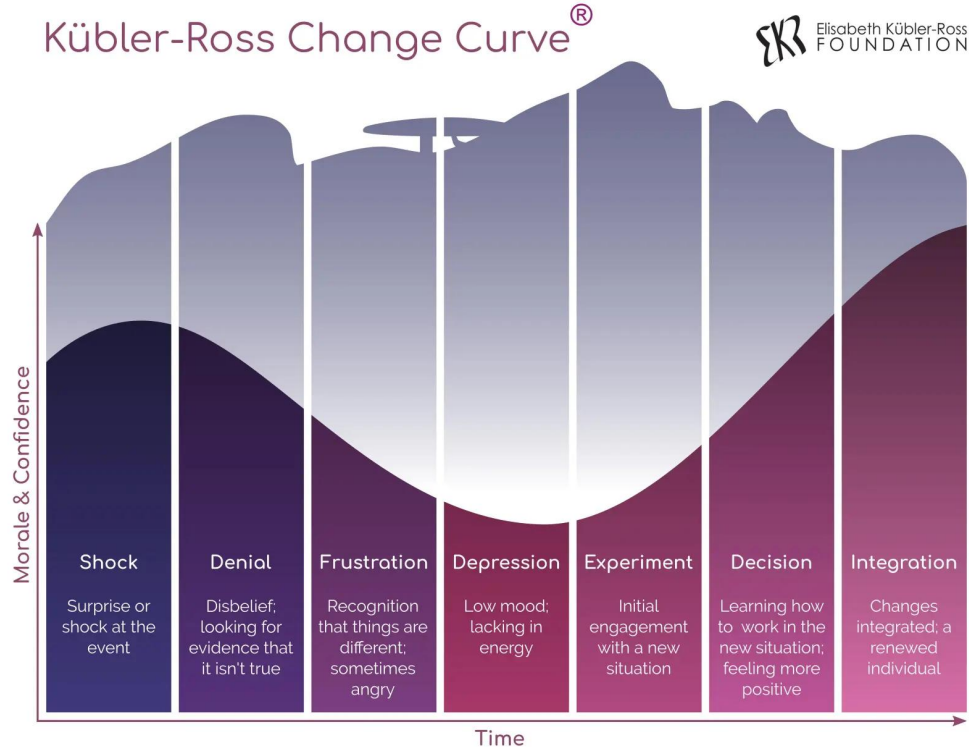
Do not fall into the engineer's trap of building everything yourself.

You may want to buy first, to experiment, until AI has gained some traction and more innovative projects with more custom requirements emerge.

Some companies offer an open source, community version that allows you to try things out on a small scale with limited functionality.

Change Management

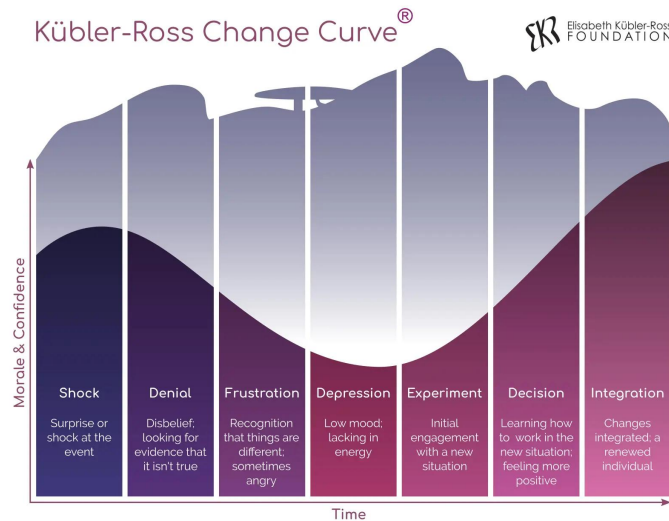
Change goes slowly, and people generally dislike change



How to accelerate change

Change goes slowly, and people generally dislike change

- Shock/Denial:
 - Create a sense of urgency
 - Explain, because generally people are reasonable.
- Frustration / Depression:
 - Involve stakeholders in the discussion. Do not only let them react.
 - Listen and observe: try to accommodate to worries, organisational pains, etc. Change may affect people's jobs.
- Experiment
 - Create a guiding coalition (bottom-up) that is capable of achieving quick wins and influencing others.
- Decision / Integration:
 - Work with priorities: make this part of the business strategy (top-down). Or (ask to) drop certain tasks, because we cannot keep doing everything.
 - Provide support, e.g. to work with new technology.
 - Provide resources as change / migrations may require additional effort.



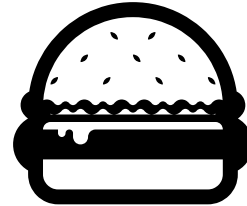
Remember, change takes time.

10-minute Break!

This will help you learn



10 minutes





What are the key AI Portfolio dimensions that influence Use Case selection? Select all that apply.



1. Ideation / prioritisation



2. Value / cost management

3. Change management

4. Build vs Buy



Build or Buy?

'buh-bAI' is a small company focusing on employee farewell eCards. They are looking to start out with AI to generate farewell messages. Should they build their own foundation model, or should they buy an OpenAI licence?

Buy

Real-world scenario



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Just Eat Takeaway

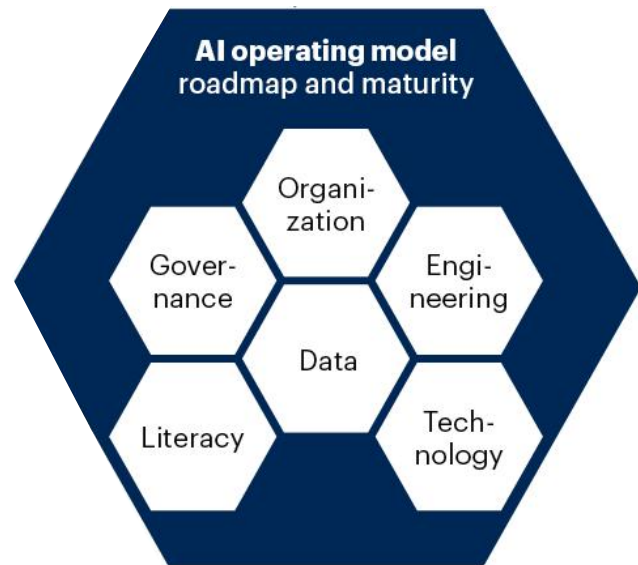
AI Operating Model (Part 1)



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The AI Operating Model in Context

- *Organisation*
- *Data*
- *Technology*
- *Engineering*
- *Governance*
- *Literacy*



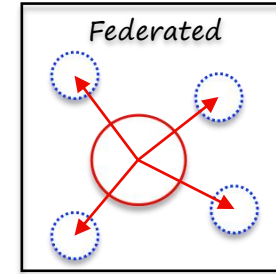
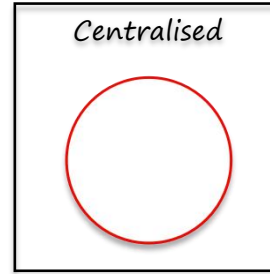
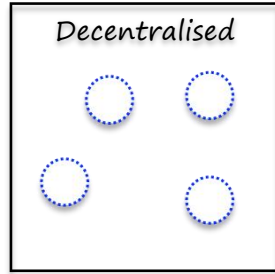
Organisation



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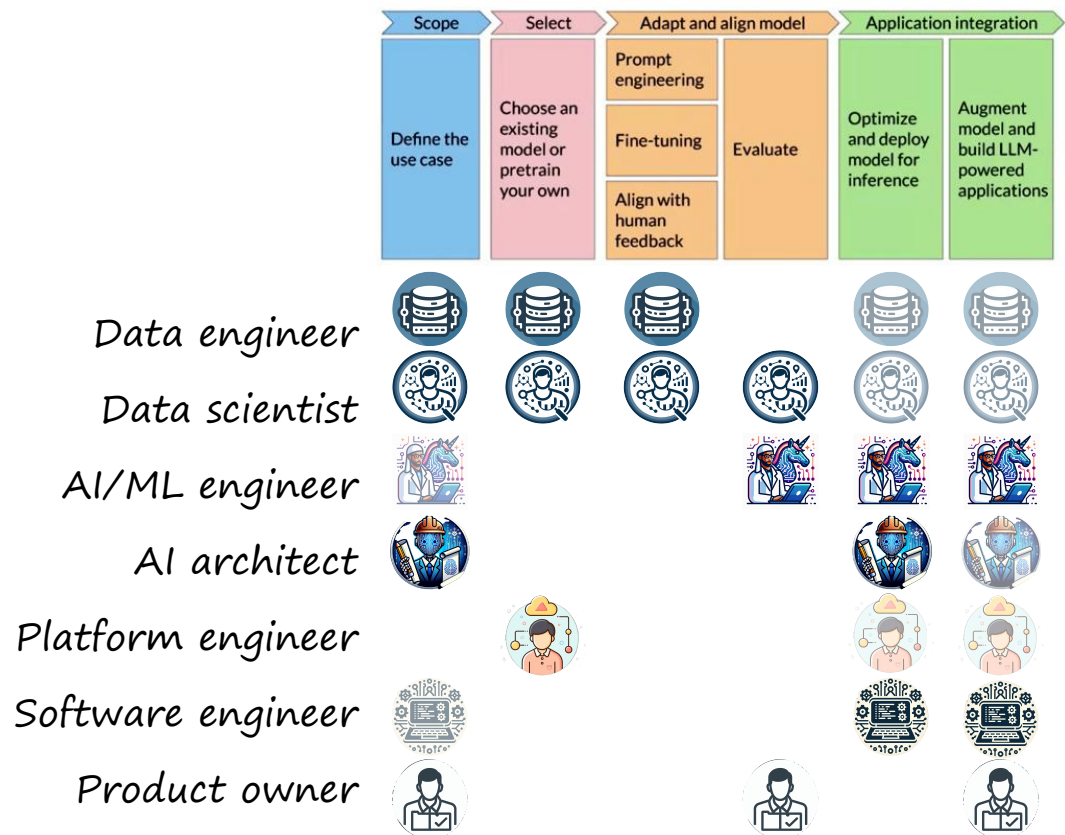
Centralised vs Decentralised

How are you going to organise for value delivery?



<i>Time-to-market</i>	Fast	Slow	Increasingly fast due to standardisation
<i>Tools / tech</i>	Risk of duplication	One set of tools / tech	Flexible' set of tools / tech
<i>Governance</i>	Challenging to control	Top-down control	Controlled through standards and guidance
<i>Communication</i>	Fast within teams	Slow comms with business	Closer ties to business
<i>Small/large orgs</i>	Can work for large organisations	Can work for small organisations	Can work for large organisations
<i>Decision making</i>	Fast, autonomous	Slower, top-down (can be positive!)	Top-down and autonomous
<i>Development opportunities</i>	Good	Fewer	Varying development opportunities: Centre-of-excellence can drive development quickly

Roles in the organisation



AI Delivery Teams

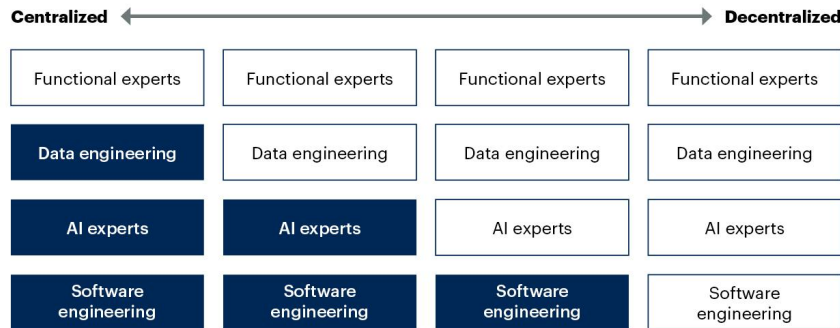
According to Gartner

Key considerations that determine where to put responsibility for AI Delivery:

- *Domain knowledge required?*
- *Existing software in the product team?*
- *Who owns the data?*
- *Where is talent currently located?*

Spectrum of Choices for the Organization of AI Delivery Teams

■ External to product team □ Part of product team



Source: Gartner
817548_C

Conway's Law

by *Melvin Conway*

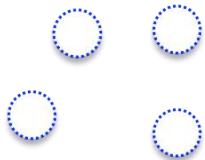
- "Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure."
- i.e. your system's development is limited by communication lines, organisational structure, etc.
 - if you have three teams working on a product, the product will consist of three steps / components.
- 'Reverse Conway Manoeuvre': Create an organisational structure and communication lines that represent how you want your system to function.



Centralised vs Decentralised

How your choice affects communication / technology / people

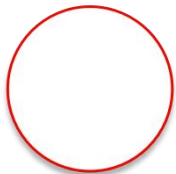
Decentralised



There could be a tendency towards a micro services architecture:

- Communication: How will your team communicate with other teams they are dependent on?
- Integration: How will your AI application integrate with software applications from other business teams (e.g. order handling software team vs. ETA prediction AI team)?
- Data: How to deal with data owned elsewhere in the company?

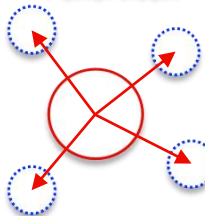
Centralised



There could be a tendency towards a monolithic architecture:

- Communication: How do you ensure business requirements and KPIs are adopted by the central AI team?
- Integration: It maybe more challenging to integrate with existing business systems. Sometimes business teams lack resources or urgency to
- Data: Data will be in one place, allowing the central team to move fast.

Federated



There could be a mismatch in priorities:

- Communication: How do you ensure business requirements and KPIs are adopted by the central AI team?
- Integration: How to integrate centrally developed AI applications with business applications?
- Data: How to deal with data owned elsewhere in the company?
- People: If you have a centralised team of AI experts, they may be blocking to projects in business pillars. Business pillars may not be up to date on recent platform developments

Team Topologies

Matthew Skelton and Manuel Pais

- *Stream-aligned teams focus around creating value; usually one, multi-functional team (i.e. front-end, back-end, etc.). Examples include a customer service chatbot team.*
- *Platform team: aims to reduce inefficiencies by creating and automating a 'golden path' traversed by the stream-aligned team, the centrally managed platform. Examples include AI Platform teams, ML(Ops) Platform teams, or Data platform teams.*
- *Enabling team: Supports the stream-aligned team with (new) knowledge and skills, but is always focused on increasing autonomy of that team. Examples include (some) data engineering teams, ML(Ops) engineering teams.*
- *Complicated Subsystem team: Some parts of the system become so complicated they may require specialised knowledge or focus from a dedicated team. Examples include security teams, FinOps teams, or data governance teams.*

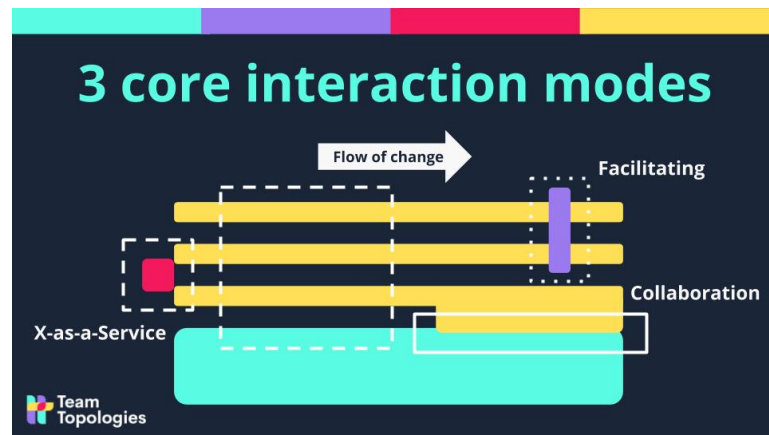


Team Topologies is a tool to create a federated organisation

Team Topologies

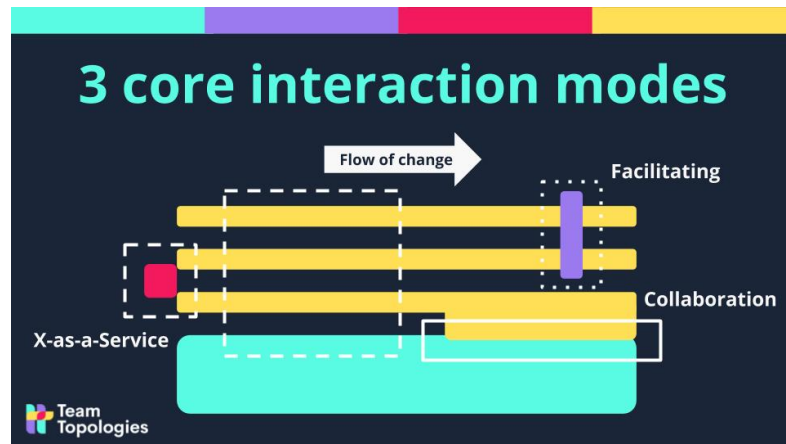
3 core interaction modes

- *X-as-a-Service* focuses on delivering a particular product or service to stream-aligned teams. Examples include (managed) Kafka clusters for inter-service communication, and
- Enablement teams can be *Facilitating* to a stream-aligned team if they lack maturity (e.g. knowledge, skills, projects, tools) or when they are unfamiliar with the centrally managed (AI) platform.
- *Collaboration*: Mature value-aligned teams are often looking to have certain platform features standardised and automated to save time, but also to increase reliability, stability, cost-efficiency, and compliance.



When to use Team Topologies

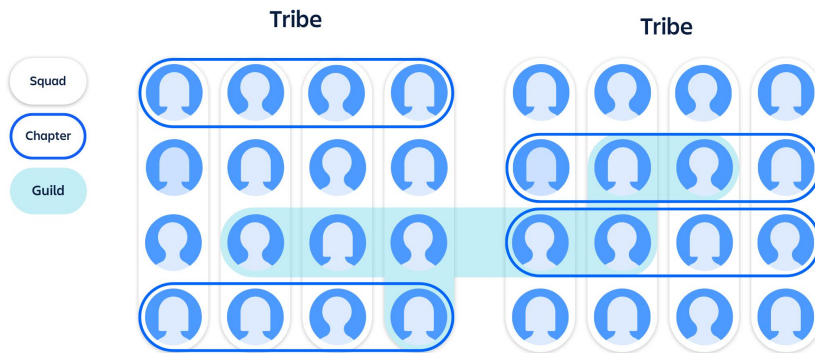
- When an organisation has a lot of duplication in systems, tools, and processes.
- When value streams share one or a few 'golden paths'.
- When value streams need autonomy to make decisions quickly and move forward quickly.
- Multiple small projects with diverse domain requirements
- When you are looking to reduce workload ('cognitive load'), for instance, due to complex legal and/or data privacy requirements



Matrix organisation

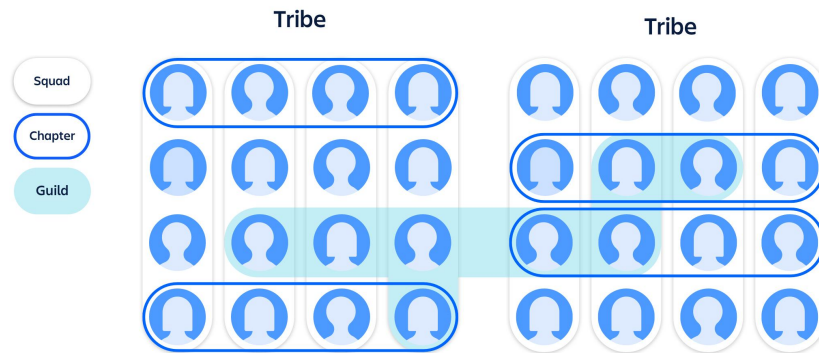
Further developed and popularised by Spotify

- Tribes focus on one feature area, like deliveries.
- Squads focus on one feature, like estimated-time-of-delivery.
- Chapters specialise around particular skill, like data science, ML engineering, or prompt engineering.
- Guilds are more transient, meet less often and focus on a particular topic, like 'Python', or 'LLMOps'.



Pros and Cons of the Spotify model

- ✓ More knowledge sharing due to existence of chapters.
- ✓ More autonomy for the squad.
- ✓ More flexible human resources, as a chapter of 4 can be part of more than 4 projects / teams.
- ✗ Balancing between multiple projects and their priorities as an individual / chapter.
- ✗ Lack of resources in a chapter may be blocking to project progress.
- ✗ Loosely tied chapters may get caught up with project work and not share any knowledge within.



Wrap-up



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Summary

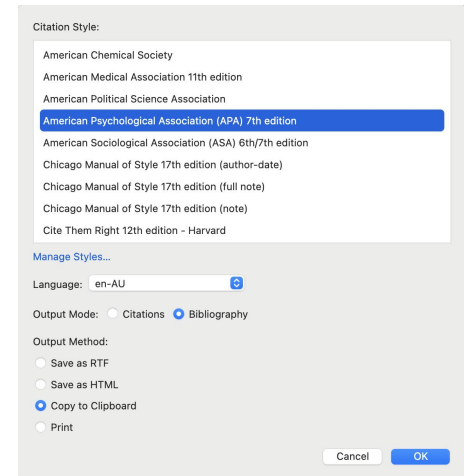
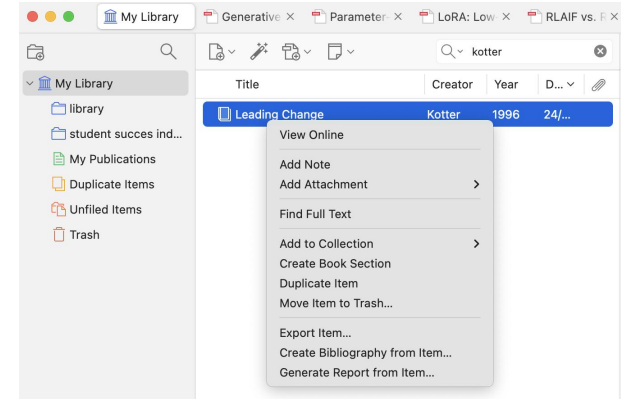
- *Alignment: Your AI Strategy needs to be aligned with the Business Strategy*
- *The AI Strategy is related to the AI Portfolio (planning) and AI Operating Model (execution)*
- *Conways' law: your organisation structure will influence your system architecture.*

Tip for your report

Ensure you refer to your resources (including GenAI)

- APA:
 - Example: Build a 'Guiding Coalition' (Kotter, 1996)
 - References section includes:
 - Kotter, J. P. (1996). *Leading Change*. Harvard Business Press.
http://books.google.nl/books?id=ib9Xzb5eFGQC&lpg=PR11&ots=Mvn9eIASZ_&dq=kotter 1996&lr&pg=PR11#v=onepage&q=kotter 1996&f=false
- IEEE
 - Example: Build a 'Guiding Coalition' [2]
 - References section includes:
 - [2] J. P. Kotter, *Leading Change*. Harvard Business Press, 1996. [Online]. Available:
http://books.google.nl/books?id=ib9Xzb5eFGQC&lpg=PR11&ots=Mvn9eIASZ_&dq=kotter 1996&lr&pg=PR11#v=onepage&q=kotter 1996&f=false

Tools: Zotero (zotero.org) and Endnote (<https://endnote.com/>). More info about their use at the [UOW Library](#).



For next lecture

Read Weidinger et al., 2021 (see Moodle)

*Data Management,
Privacy, and Ethics*

References and Resources

- Bhati, R., den Hamer, P., Shubhangi, V., & Souparna, P. (2024, July 24). *These 6 Elements Form the Foundation of Your AI Operating Model*. Gartner. <https://www.gartner.com/en>
- den Hamer, P., & Bhati, R. (2024, April 23). *The Pillars of a Successful Artificial Intelligence Strategy*. Gartner. <https://www.gartner.com/en>
- Iusztin, P., & Labonne, M. (2024). *LLM Engineer's Handbook* (1st ed.). Packt.
- Jaffri, A., Clendaniel, S., & Bhati, R. (2024, November 26). *Adopt a Hybrid Delivery Model to Mirror Your AI Maturity*. Gartner. <https://www.gartner.com/en>
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http://books.google.nl/books?id=ib9Xzb5eFGQC&pg=PR11&ots=Mvn9elASZ_&dq=kotter 1996&lr&pg=PR11#v=onepage&q=kotter 1996&f=false
- Sicular, S., Ellio, B., Hare, J., & Andrews, W. (2023, October 13). *Become an AI-First Organization: 5 Critical AI Adoption Phases*. Gartner. <https://www.gartner.com/en>

Questions



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